

AI IN GENOMICS - FINAL PROJECT

Interpretable Features vs. Protein Language Models

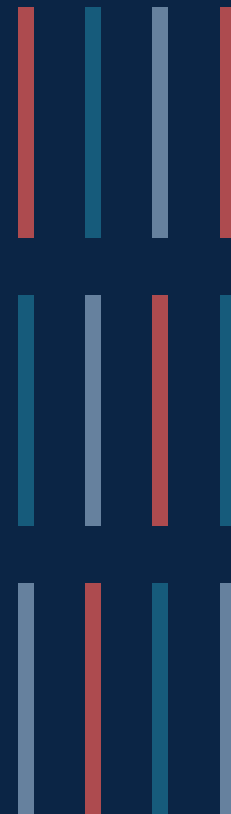
*Comparing handcrafted features and ESM-2 embeddings for missense
variant pathogenicity classification*

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A patient walks into a clinic.

Their DNA is sequenced. The lab finds **hundreds of differences** from the reference human genome.

Most are harmless. A few might cause disease. The doctor needs to know **which is which**.

Today, roughly **40% are reported as 'uncertain'** - no answer, no action.

THE BOTTLENECK

~40%

*of clinically reported variants are
'variants of uncertain significance'*

That's the gap we want AI to close.

Zooming in

Which letter changes break the protein?

We focus on **missense variants** - changes that swap one amino acid in a protein for another.

Some break the protein **(pathogenic)**.

Others are tolerated **(benign)**.

Telling them apart *from sequence alone* is the open problem.

...K-L-V-A-G-T-R-N...

Reference

...K-L-V-**C**-G-T-R-N...

Pathogenic - protein breaks

...K-L-V-**I**-G-T-R-N...

Benign - protein still works

Two questions

Q1

How much signal lives in simple, interpretable features?

Identity of the amino acids, substitution score, local composition, position. 62 dimensions.

Q2

Does ESM-2 actually help when tested on new genes?

Evaluated on entire genes held out from training - the clinically realistic setting.

The tradeoff: interpretability vs. compute vs. performance vs. generalization.

What we built, and what makes it different

WHAT WE BUILT

Four classifiers

trained on 193K labelled variants - then compared head-to-head against AlphaMissense and CADD on the same held-out test set.

WHY IT'S DIFFERENT

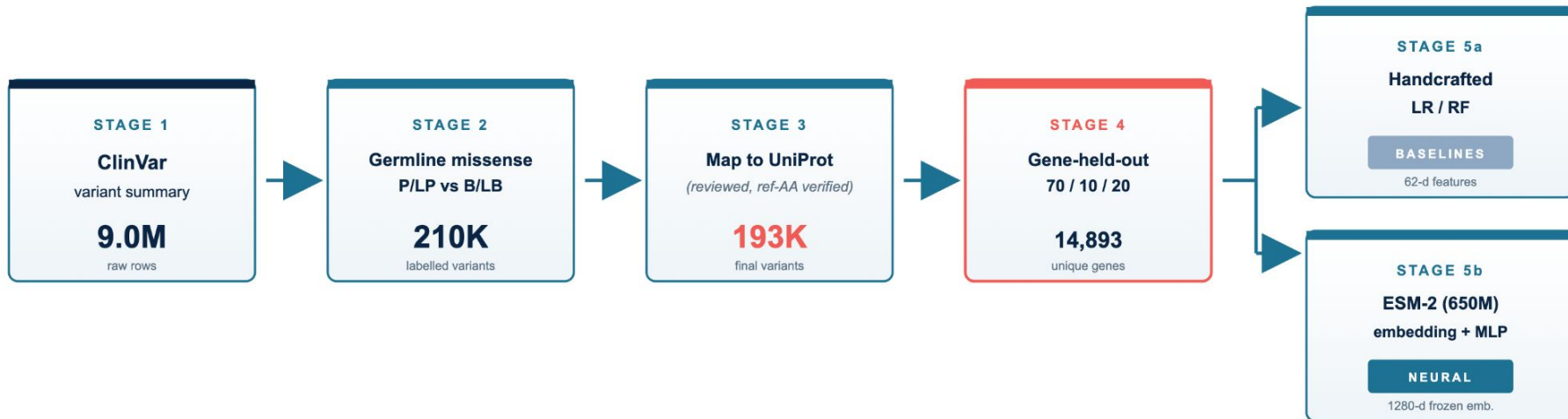
- 1 Strict gene-disjoint test - no variant from a training gene appears at test time.
- 2 Feature-family ablation directly quantifies the interpretability-vs-performance tradeoff.
- 3 We empirically test the UR50 pretraining contamination concern - not just flag it.

ACT 2 - THE SETUP

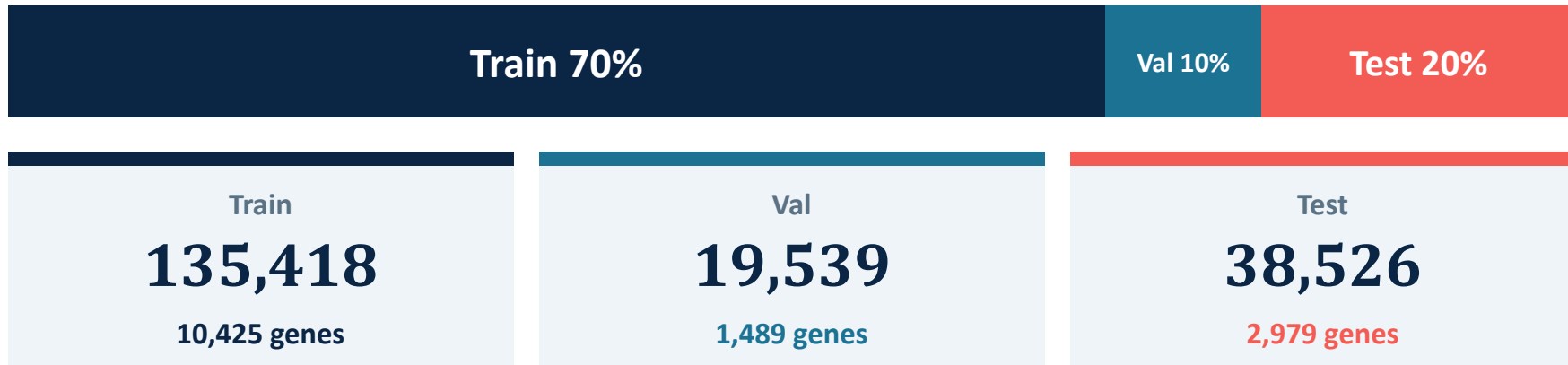
Step 1: assembling clean data

From 9 million rows to 193K usable variants

Data pipeline: from ClinVar to four trained classifiers



We hold out entire genes, not just variants



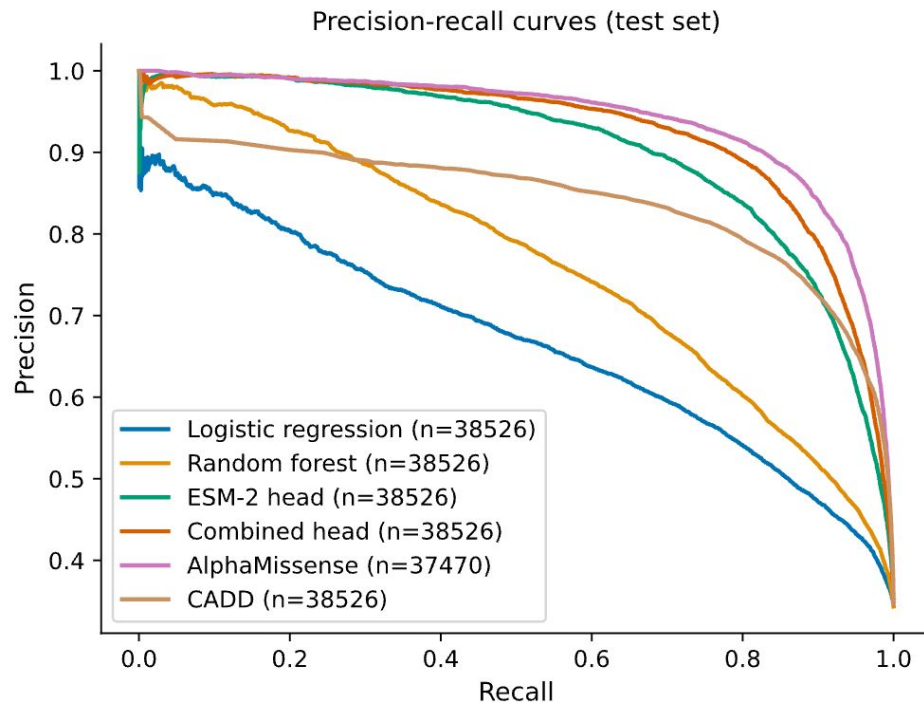
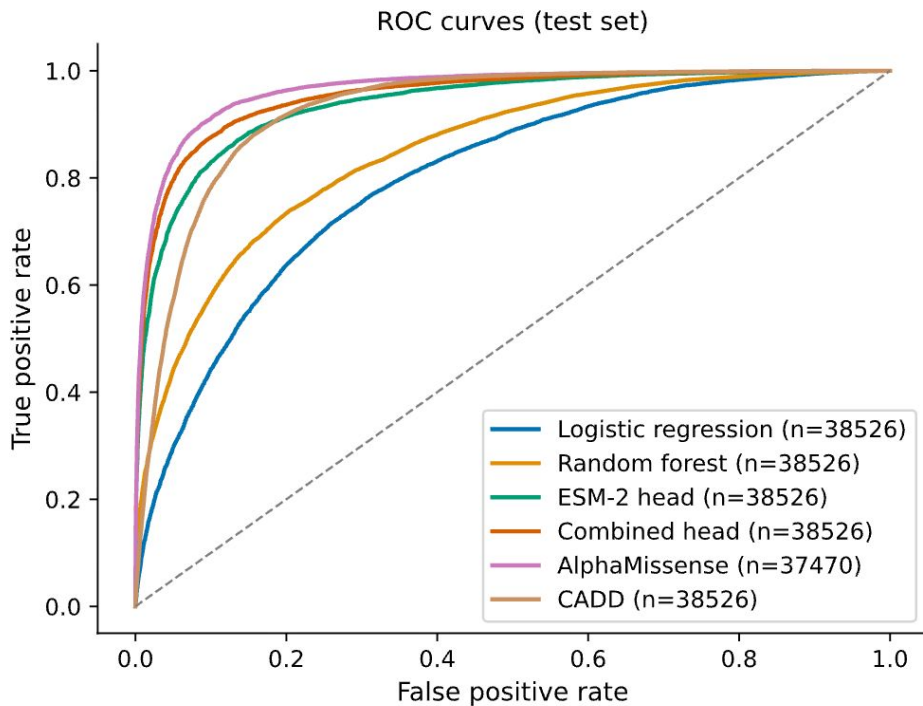
Why it matters: if any gene appears in both train and test, the model can cheat by memorizing gene-specific patterns. By splitting on whole genes, we force every test gene to be brand new.

Four of ours, two outside references

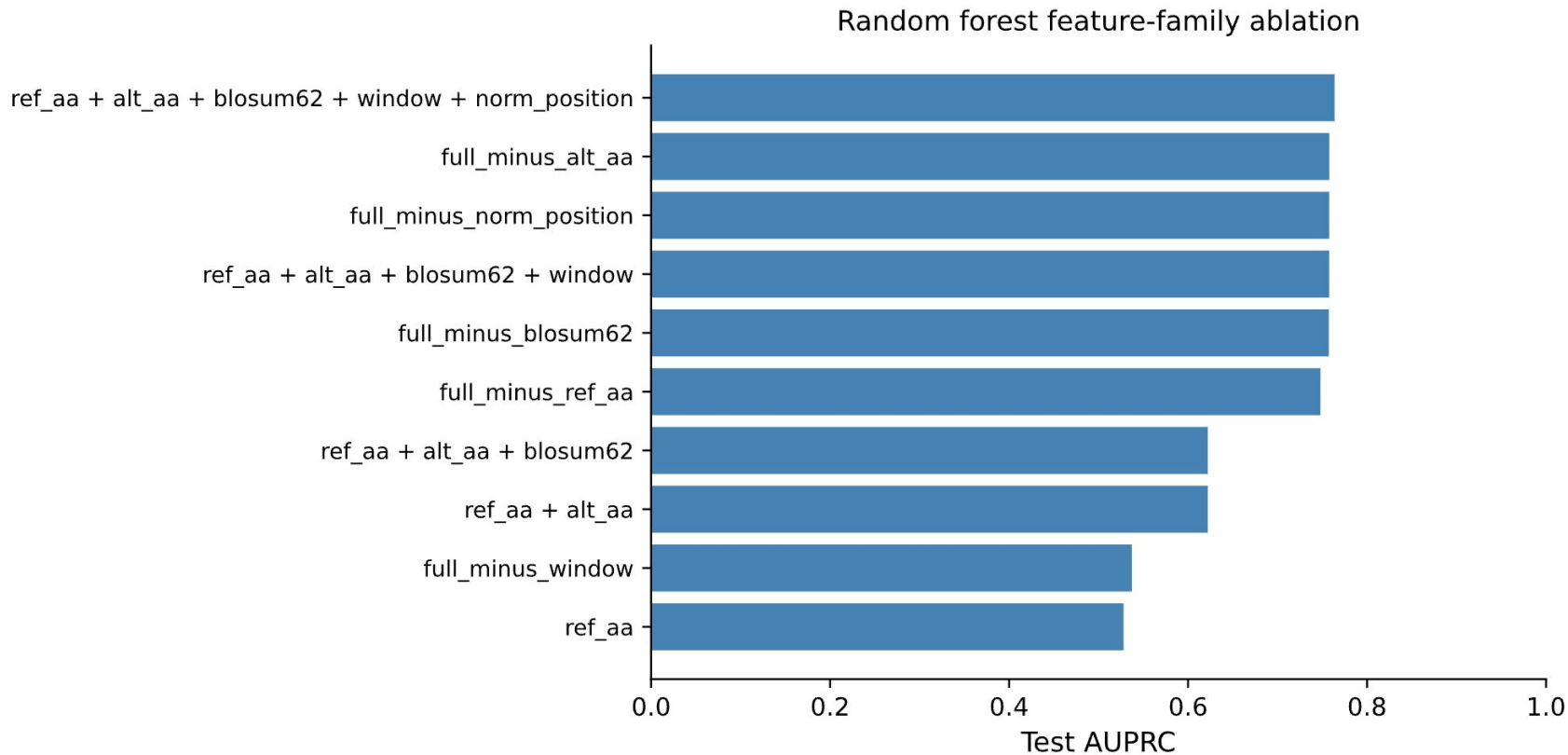
MODEL	INPUT	ARCHITECTURE	
Logistic regression	62-d handcrafted	L2 LR, class-balanced	BASILINE
Random forest	62-d handcrafted	500 trees, tuned	BASILINE
ESM-2 head	1280-d frozen emb.	MLP, AdamW	NEURAL
Combined head	1342-d (concat)	MLP, wider input	NEURAL

Outside references: **AlphaMissense** and **CADD** - we score them on the same test variants to know where the field stands.

ESM-2 closes most of the gap to AlphaMissense

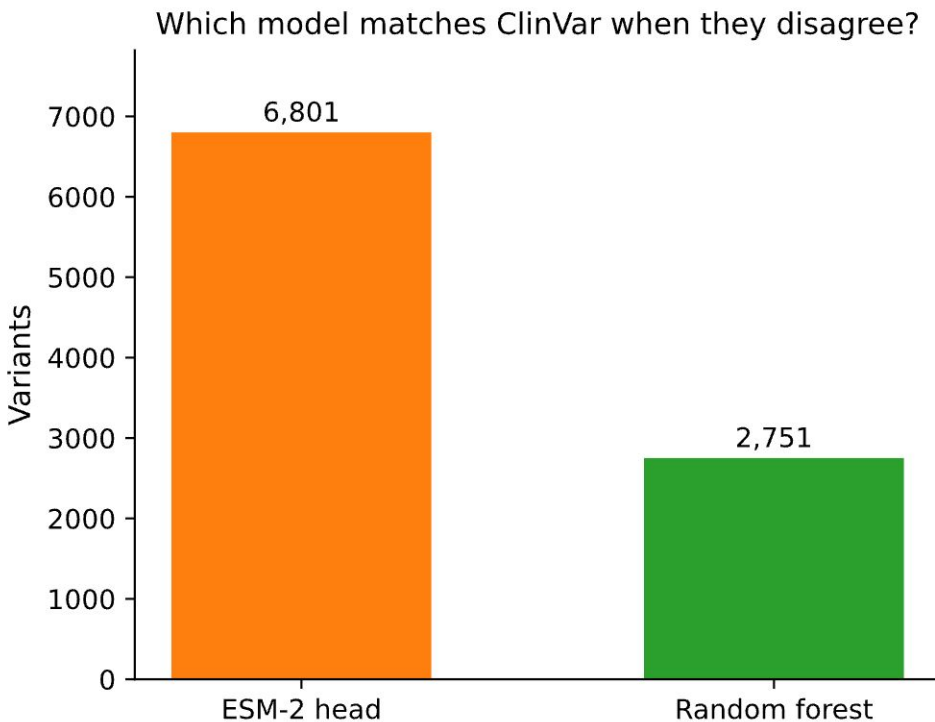
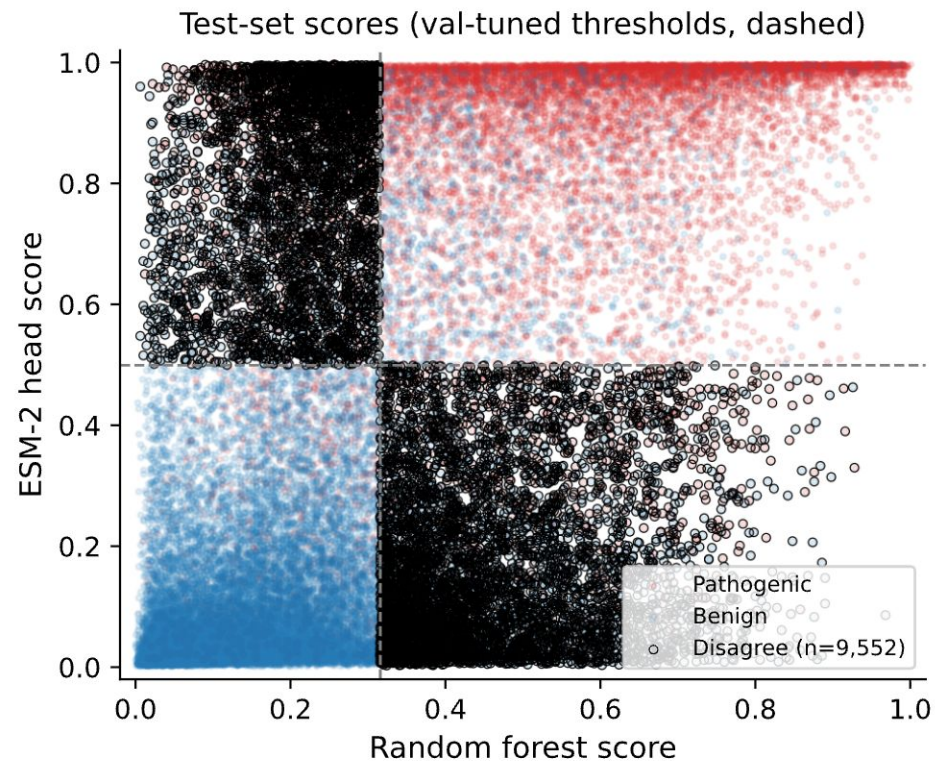


Simple features get you 81% of the way there

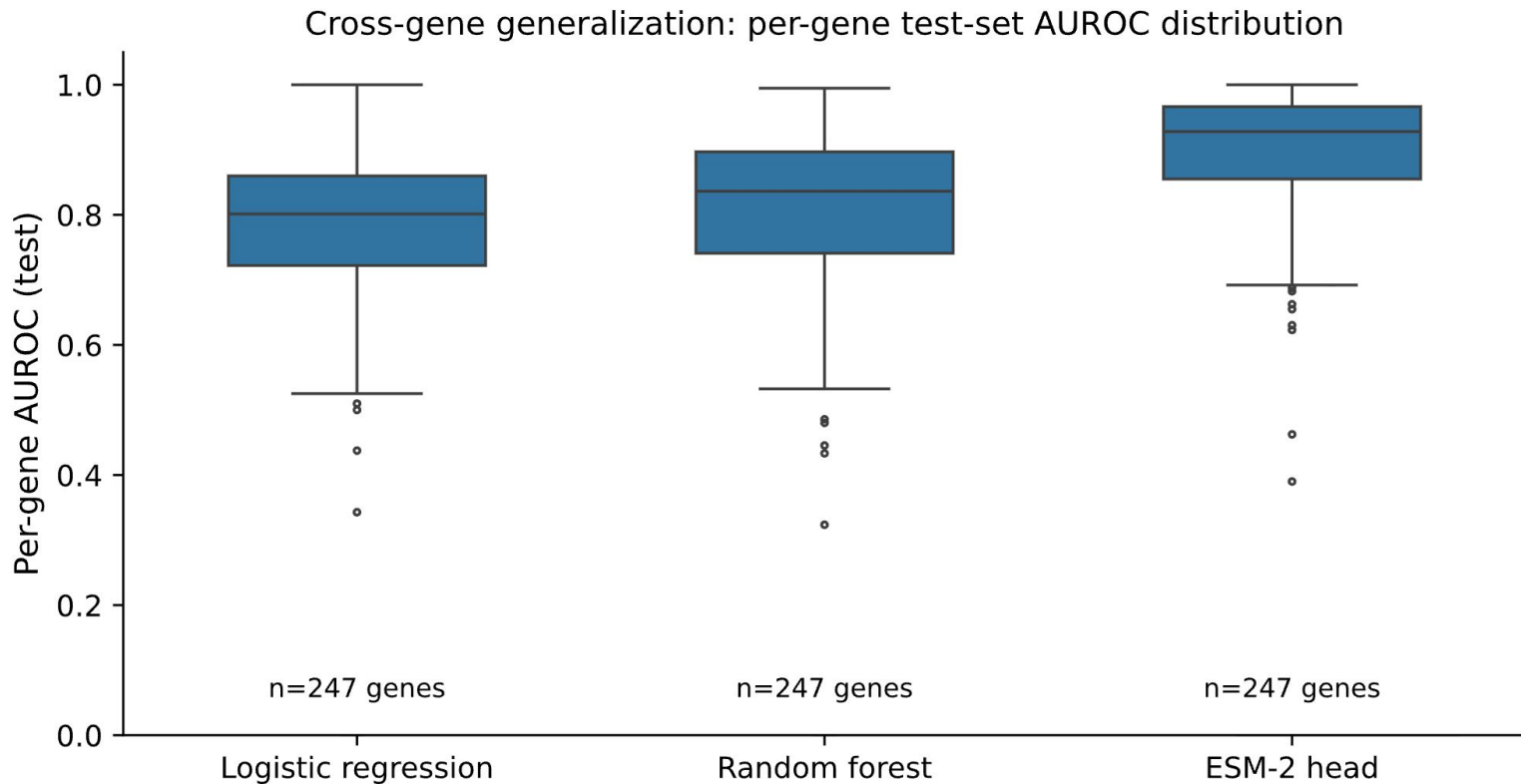


Looking inside the gap

When RF and ESM-2 conflict, ESM-2 wins more often

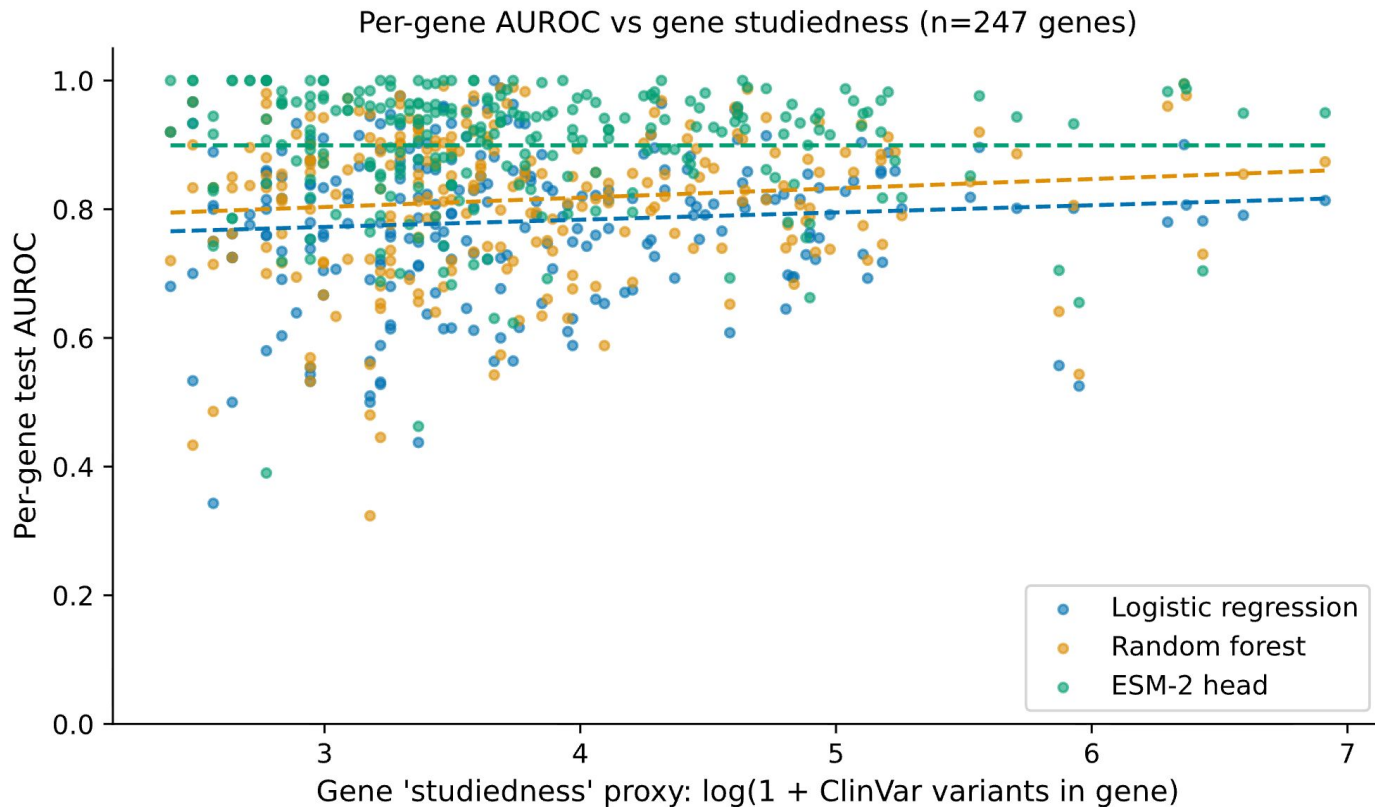


ESM-2 generalizes evenly across genes



But wait...

Is ESM-2 just winning on well-studied genes?



Three things we learned, plus what's next

1

Simple features carry real signal. A 62-d handcrafted vector reaches AUROC 0.850 - and just three of those features get you to 81% of that.

2

ESM-2 truly generalizes. +0.089 AUROC over the random forest on held-out genes ($p < 0.001$). The advantage isn't from UR50 contamination.

3

Combining is best. Concatenating handcrafted features with ESM-2 embeddings gives another +0.015 - they carry complementary information.

LIMITS ClinVar bias, optimistic references

NEXT Fine-tune ESM-2 end-to-end on ClinVar

Thank You!

